VU Medical Zone

CHE 201 Current Final terms Fall 2020

M.C.Q

1. Thermodynamics studies the relationships between the _____of a system. **Macroscopic properties**

2. Two systems in thermal equilibrium with each other have the same_____. **Temperature**

3. For a system in mechanical equilibrium, the pressure throughout the system is_____. **Uniform**

4. According to Charles law the volume of a fixed amount of gas, as constant pressure, is proportional to the_____ . **Absolute temperature**

5. Calorimetry is_____. **Measurement of Enthalpy**

6. A measurement of the relative amounts of products and reactants present in a reaction at a given time is called _____. **Reaction quotient**

7. Number of moles of solute per liter of the solution_____. **Molarity**

8. _____are liquids or solids with a gas dispersed into them. **Foams**

9. The interactions between the solute particles and the solvent molecules is called_____. **Solvation**

10. _____ is often used as a way to purify a compound. **Precipitation**

11. In fractional distillation fractionating column is packed with glass beads to give_____. **Maximum possible surface area for vapor to condense on**

12. Continuous Flow is a type of assay used for determining initial value and_____. **Inhibition values**

13. The rate at which the reagents are first brought together is called_____. **Initial rate**

14. The relaxation methods can be used to determine the_____. **Rate of the reaction**

15. A first-order reaction is a reaction that proceeds at a rate that depends on_____. **One reactant concentration**

16. Boiling point of Ethanol is _____. **78.5°C**

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17. The number of particles involved in an elementary step is called_____. **Molecularity**

18. The fan in a 'frost free' home freezer moves air around inside the freezer to_____. **Sublimate**

19. Extensive properties depend on the _____ of the system. **Parts**

20. A non-spontaneous is also known as _____. **Endergonic reaction**

21. _____ influences the physical properties of liquids. **Intermolecular forces**

Subjective

Qno1: What is the effect of catalyst on the activation energy? 2 marks

If a catalyst is added to the reaction, the activation energy is lowered because a lower energy transition state is formed.

Qno2: What is pseudo first order reaction? 2 marks

Under certain conditions, the second order kinetics can be well approximated as first order kinetics. These Pseudo first order reactions greatly simplify quantifying the reaction dynamics.

Qno3: What is elementary step of reaction mechanism? 3 marks

The elementary steps used for mechanisms of chain reactions can be grouped into the following categories:

- Initiation step
- Chain propagation steps
- Chain branching steps
- Chain inhibition steps
- Chain termination steps

Qno4: What is the mechanism of stopped flow mechanism? 5 marks

Stopped Flow

• The stopped-flow technique allows for the evaluation of solution-based kinetics on milliseconds timescale with a very small volume of reactants used.

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Mechanism

• This technique involves two reactants held in separate reservoirs that are prevented from freely flowing by syringe pumps.

• The reaction is initiated by depressing the reactant syringes, and thus releasing the reactants into the connecting "mixing chamber" where the solutions are mixed.

• The reaction is monitored by observing the change in absorbance of the reaction solution as a function of time.

• As the reaction progresses it fills the "stop syringe" which then expands until it hits a block at the point when the reaction has reached a continuous flow rate, thereby stopping the flow and the reaction, and thus allowing the researcher to calculate the exact initial rate of reaction.

Qno5: Write down the Limitations of Thermodynamics. 5 marks

Limits of Thermodynamics in Reaction mechanisms:

• Thermodynamics is a powerful approach toward understanding chemical reactions, but only provides part of the picture. Specifically:

• Thermodynamics only points the way

• Thermodynamics says nothing about how long it takes to get there

• The stoichiometric equation for the reaction says nothing about its mechanism

Thermodynamics only points the way

• Chemical change is driven by the tendency of atoms and molecules to rearrange them in a way that results in the maximum possible dispersion of thermal energy into the world.

• The observable quantity that measures this spreading and sharing of energy is the free energy of the system.

• Three reactions that look alike, but are different:

Consider, for example, the gas-phase formation reactions of the hydrogen halides from the elements.

The thermodynamics of these reactions are all similar (they are all highly exothermic), but their dynamics (their kinetics and mechanisms) could not be more different.

Qno6: Define ppm. 2 marks

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We get (1 g solute)/(I000000 g solution). Because both the solute and the solution are both now expressed in terms of grams, it could now be said that the solute concentration is 1 part per million (ppm).

Qno7: Write down the practical applications of Thermodynamics. 5 marks

Another application of this law is refrigerators and heat pumps based on the Reversed Carnot Cycle. If you want to move heat from a body at a lower temperature to a body at a higher temperature, then you have to supply external work. In the original Carnot Cycle, heat produces work while in the Reversed Carnot Cycle work is provided to transfer heat from lower temperature reservoir to a higher temperature reservoir.

Removing heat from the food items in the refrigerator and throwing it away to the higher temperature atmosphere doesn't happen automatically. We need to supply external work via the compressor to make this happen in the refrigerator.

Air conditioner and heat pump follow the similar law of thermodynamics. The air conditioner removes heat from the room and maintains it at a lower temperature by throwing the absorbed heat into the atmosphere. The heat pump absorbs heat from the atmosphere and supplies it to the room which is cooler in winters.

Qno8: Write down the methods for the measurement of reaction rate. 3 marks

There are following methods used for the measurement:

1. Continuous Flow

2. Methods for measuring concentration

3. Rate v/s Concentration Proportionalities

4. Relaxation method

5. Spectrometry

6. Stopped Flow

Qno9: Define the Process of Precipitation. 5 marks

Precipitation is the process of a compound coming out of solution. It is the opposite of dissolution or solvation. In dissolution, the solute particles separate from each other and are surrounded by solvent molecules. In precipitation, the solute particles find each other and form a solid together. This solid is called the precipitate or sometimes abbreviated "ppt".

Qno9: Define Azeotrope. 3 marks

An azeotrope is a mixture that exhibits the same concentration in the vapor phase and the liquid phase. This is in contrast to ideal solutions with one component typically more volatile than the other; this is the reason we use distillation to separate materials.

Qno10: Define Brine, syrup, saline and seltzer. 5 marks

A solution of water and sugar is called syrup.
A solution of sodium chloride (common table salt) in water is called brine.
A sterilized specific concentration (0.15 molar) of sodium chloride in water is called saline.
A solution of carbon dioxide in water is called seltzer, and a solution of ammonia gas in water is called ammonia water.

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2 marks

Qno1: Write down the methods for the measurement of the reaction rate?

These are the methods for the measurement of the reaction rate

1. Continuous Flow

2. Methods for measuring concentration

3. Rate vs. Concentration Proportionalities

4. Relaxation Methods

5. Spectrophotometry

6. Stopped Flow

Qno2: On addition of solute what happen to vapor pressure according to Raoult's Law?

Adding a solute lowers vapor pressure because the additional solute particles will fill the gaps between the solvent particles and take up space.

• This means less of the solvent will be on the surface and less will be able to break free to enter the gas phase, resulting in a lower vapor pressure.

• There are two ways of explaining why Raoult's Law works - a simple visual way, and a more sophisticated way based on entropy.

Qno3: Define Charles Law.

Jacques Charles discovered that the volume of a fixed amount of gas, as constant pressure, is proportional to the absolute temperature.

Qno4: Define catalysis.

Catalysis is the process of increasing the rate of a chemical reaction by adding a substance known as a catalyst. Catalysts are not consumed in the catalyzed reaction but can act repeatedly.

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Qno5: Define solubility.

The solubility of a solution is a measure of how much of the solute can be dissolved into the solvent.

Qno6: What is reaction mechanism?

Chemical reactions are studied in terms of reaction rate, reaction orders with respect to reactants, differential and integrated rate laws of chemical reactions, and activation energy. This type of studies is usually called chemical kinetics.

3 marks

Qno1: Write down the effect of catalyst on activation energy.

A catalyst provides a reaction route with lower activation energy. There are almost 30,000 times more molecules which can react in the presence of the catalyst compared to having no catalyst (using our assumptions about the activation energies).

Qno2: How to measure the viscosity by Ostwald method?

• Another more advanced design of measuring viscosity known as the Ostwald Viscometer that is much more accurate than dropping a metal ball.

• An Ostwald Viscometer consists of two reservoir bulbs and a capillary tube.

• The viscometer is filled with liquid until the liquid reaches the mark A with the aid of a pipette to accurately measure out the volume of needed liquid.

The viscometer is then put into a water bath which equilibrates the temperature of the test liquid.

• As noted before, the equilibration is important to maintain a constant temperature as to not affect the viscosity otherwise.

• The liquid is then drawn through the side 2 of the U-tube by use of suction and lastly, the flow is time between marks C and B.

• The viscosity is calculated with Equation nu=Kt Where K is the value of a liquid with known viscosity and density such as water. Once the value of K is known, the viscosity can be determined by measuring the amount of time the test liquid flows between the two graduated marks.

Qno3: Define critical temperature, pressure and volume.

The Critical Volume: The volume occupied by a unit mass of a gas or vapors in its critical states.

The Critical Pressure: The pressure required to liquefy a gas at its critical temperature

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The Critical Temperature: The temperature at and above which vapors of the substance cannot be liquefied, no matter how much pressure is applied.

Qno4: Define Solvation.

Dissolution means the process of dissolving or forming a solution. When dissolution happens, the solute separates into ions or molecules, and each ion or molecule is surrounded by molecules of solvent. The interaction between the solute particles and the solvent molecules is called solvation. A solvated ion or molecule is surrounded by solvent.

Qno5: Write down the Elementary steps.

The elementary steps used for mechanisms of chain reactions can be grouped into the following categories:

- Initiation step
- Chain propagation steps
- Chain branching steps
- Chain inhibition steps
- Chain termination steps

Qno6: Define open, close and isolated systems.

Open system is one where transfer of matter between system and surroundings can occur.

Closed system is one where no transfer of matter can occur between system and surroundings.

Isolated system is one that does not interact in any way with its surroundings.

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5 marks

Qno1: Write a note on precipitation.

Precipitation is the process of a compound coming out of solution. It is the opposite of dissolution or solvation. In dissolution, the solute particles separate from each other and are surrounded by solvent molecules. In precipitation, the solute particles find each other and form a solid together. This solid is called the precipitate or sometimes abbreviated "ppt".

Qno2: Explain the mechanism of stopped flow mechanism.

Mechanism

• This technique involves two reactants held in separate reservoirs that are prevented from freely flowing by syringe pumps.

• The reaction is initiated by depressing the reactant syringes, and thus releasing the reactants into the connecting "mixing chamber" where the solutions are mixed.

• The reaction is monitored by observing the change in absorbance of the reaction solution as a function of time.

• As the reaction progresses it fills the “stop syringe” which then expands until it hits a block at the point when the reaction has reached a continuous flow rate, thereby stopping the flow and the reaction, and thus allowing the researcher to calculate the exact initial rate of reaction.

• The Stopped Flow technique works because within milliseconds of combining the two reactants the absorbance can be read.

• In addition, the stop syringe assures for a steady rate of flow pas the spectrophotometer so that reactants are being added to solution and forming products at a consistent rate.

Qno3: Explain Limitations of thermodynamics.

Limits of Thermodynamics in Reaction mechanisms 1

• Thermodynamics is a powerful approach toward understanding chemical reactions, but only provides part of the picture. Specifically:

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Limits of Thermodynamics in Reaction mechanisms 2

• Three reactions that look alike, but are different: Consider, for example, the gas-phase formation reactions of the hydrogen halides from the elements. The thermodynamics of these reactions are all similar (they are all highly exothermic), but their dynamics (their kinetics and mechanisms) could not be more different.

Qno4: Write down the applications of kinetics studies.

Kinetic studies are important in understanding reactions, and they also have practical implications.

• For example, in industry, reactions are conducted in reactors in which compounds are mixed together, possibly heated and stirred for a while, and then moved to the next phase of the process.

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• It is important to know how long to hold the reaction at one stage before moving on, to make sure that reaction has finished before starting the next one.

• By understanding how a reaction takes place, many processes can be improved.

• For example, if it is known that a particular intermediate is involved in a reaction, the use of conditions (such as certain solvents) that are incompatible with that intermediate might be avoided.

• In addition, reagents might be added that would make certain steps in the reaction happen more easily.

• Not only are kinetic studies important in industry, but they are also used to understand biological processes, especially enzyme-catalyzed reactions.

• They also play a role in environmental and atmospheric chemistry, as part of an effort to understand a variety of issues ranging from the fate of prescription pharmaceuticals in wastewater to the cascade of reactions involved in the ozone cycle.

10 marks

Qno1: Explain 1st, 2nd and 3rd order of a reaction.

First Order Reactions

A first-order reaction is a reaction that proceeds at a rate that depends linearly on only one reactant concentration.

Second Order Reactions 1

Many important biological reactions, such as the formation of double stranded DNA from two complementary strands, can be described using second order kinetics. In a second order reaction, the sum of the exponents in the rate law is equal to two.

• The two most common forms of second order reactions will be discussed in detail in this section.

Third Order Reaction

A reaction is said to be of third order if the rate is determined by the variation of three concentration terms. In other words, the minimum number of molecules necessary for the reaction to take place is three.

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